

		Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Department: AI-ML/AI-RO/AI/AI-DS/CSE/MICRO/SAP/QUICK/ORACLE/IT/AERO		
Subject Name	PROBABILITY, STATISTICS AND NUMERICAL METHODS	A.Y	2025/2026	
Subject Code	303191251	Semester	4th	
Tutorial-5				
Sr No	Question	COs	B.T	Competence
1	Construct the forward difference table for the data: $x = 0, 1, 2, 3, 4$ and $f(x) = 1, 2, 4, 7, 11$.	5	2	Understand
2	Using Newton's Forward Interpolation Formula , find $f(2.5)$ for the data: $x = 0, 1, 2, 3$; $f(x) = 1, 2, 4, 7$.	5	3	Apply
3	Using Newton's Backward Interpolation Formula , estimate $f(5.5)$ using the table: $x = 3, 4, 5, 6$ and $f(x) = 27, 64, 125, 216$.	5	3	Apply
4	Using Newton's Divided Difference Formula , find $f(4)$ for the data: $(2, 3), (3, 6), (5, 12), (6, 15)$.	5	4	Analyze
5	Find the Lagrange interpolation polynomial for the points: $(1, 1), (2, 4), (3, 9)$. Hence evaluate $f(2.5)$.	5	5	Evaluate
6	Show that $\Delta \nabla f(x) = f(x+1) - 2f(x) + f(x-1)$ using properties of finite difference operators .	5	2	Understand
7	For the following data: $x = 10, 20, 30, 40$ and $y = 0.173, 0.342, 0.512, 0.683$, find $f(25)$ using divided differences .	5	4	Analyze
8	Using Lagrange interpolation , find the polynomial passing through $(0,1), (1,3), (3,13)$. Then compute $f(2)$.	5	6	Create
9	Verify that the forward and backward interpolation formulas give the same result for symmetric data: $x = -1, 0, 1$ and $f(x) = 1, 0, 1$.	5	5	Evaluate
10	The population of a town for years 2000, 2005, 2010, 2015 was 50,000; 58,000; 70,000; 85,000. Use an	5	4	Analyze

	appropriate interpolation method to estimate the population in the year 2012.			
--	--	--	--	--